

DualTower: A Minimal Demo Paper for PaperPilot

Synthetic demo paper generated by demo/make_demo_pdf.py · free to copy, modify and redistribute (MIT).

Abstract

This demo paper is synthetic and exists only to exercise the reading pipeline.
We propose a dual-tower retrieval architecture for composed image retrieval.
The method reaches 87.3% accuracy on the public benchmark.

1 Introduction

Composed image retrieval takes a reference image plus a modification text as input.
Existing single-tower models entangle visual and textual signals in one encoder.
This demo paper serves as a fixed, freely redistributable example document.

2 Method

We propose a dual-tower retrieval architecture that encodes images and text separately.
The two towers are aligned with a contrastive learning objective over 1,024 batches.
A lightweight reranker then fuses the two similarity scores.

3 Results

The method reaches 87.3% accuracy on the public benchmark.
Table 1 reports the accuracy of each variant on the test set.

Model	Accuracy	Note
Single tower	84.1	baseline
Dual tower (ours)	87.3	+3.2
Dual tower + rerank	88.0	+3.9

Ablation shows the reranker contributes 0.7 accuracy points.

4 Conclusion

We presented a dual-tower retrieval architecture with a reranker.
The demo document is intentionally short so the pipeline finishes quickly.